

S4 Business Thinking ERP — Complete Enterprise Build Roadmap

সহজ বাংলায় বুঝুন

এই roadmap follow করলে আপনি ধাপে ধাপে একটি full enterprise ERP software build করতে পারবেন।

এই ERP হবে:

- Online + Offline
- Mobile + Desktop
- SaaS + Multi-Branch
- Inventory + POS + Accounting
- License System সহ
- হাজার হাজার user support-ready

এই roadmap এর সবচেয়ে important জিনিস:

- কোনটার পরে কোনটা build করবেন
- কোনটা আগে করলে future এ সমস্যা কম হবে
- Claude কে কীভাবে instruction দিবেন
- কিছু missing না রাখা

সবচেয়ে important কথা

একবারে পুরো ERP generate করতে যাবেন না।

এভাবে build করবেন:

1. Architecture
2. Database
3. Auth
4. Tenant + Branch
5. License
6. Inventory
7. Purchase
8. Sales + POS
9. Accounting
10. Offline System
11. Electron Desktop
12. Reports

- 13. Deployment
- 14. Security
- 15. Final Testing

এটাই real enterprise workflow.

Goal

এই roadmap অনুযায়ী কাজ করলে আপনি phase-by-phase একটি full production-grade enterprise ERP ecosystem build করতে পারবেন যেখানে online + offline + desktop + SaaS + multi-branch + multi-tenant + licensing system থাকবে।

এই roadmap এর purpose:

- কিছু miss না হওয়া
 - AI context loss কমানো
 - phase-by-phase stable build করা
 - production-ready architecture maintain করা
 - future scaling নিশ্চিত করা
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MASTER RULES

সবসময় এই **rules follow** করবেন

1. কখনো full ERP একবারে generate করতে বলবেন না
 2. সবসময় phase-by-phase build করবেন
 3. প্রতিটি phase শেষে:
 4. review
 5. run
 6. testing
 7. bug fixing করবেন
 8. সবসময় multi-file architecture maintain করবেন
 9. সবসময় production-ready code চাইবেন
 10. সবসময় backend + database + API + frontend together build করবেন
 11. mock/demo/fake logic reject করবেন
 12. সবসময় migration + setup + test instructions চাইবেন
 13. প্রতিটি phase Git commit করবেন
 14. প্রতিটি module integration test করবেন
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PROJECT FOUNDATION

STEP 1 — Create Claude Project

Claude এ:

1. New Project
2. ERP skill enable করুন
3. Project name:

S4 Business Thinking ERP

STEP 2 — FIRST MASTER PROMPT

Claude chat এ এই prompt দিবেন:

Design the complete production-grade ERP architecture first.

Do not generate frontend UI first.

First generate: 1. Full scalable folder structure 2. Multi-tenant database architecture 3. Branch architecture 4. Inventory transaction flow 5. Accounting transaction flow 6. Authentication architecture 7. Offline-online synchronization architecture 8. API architecture 9. Deployment architecture 10. Security architecture 11. Licensing architecture 12. SaaS architecture 13. Electron desktop architecture 14. PWA architecture 15. Backup and disaster recovery architecture

Then wait for approval before generating implementation code.

STEP 3 — REVIEW ARCHITECTURE

Check করুন:

1. PostgreSQL আছে?
2. Django + DRF আছে?
3. Redis + Celery আছে?
4. JWT আছে?
5. Multi-tenant আছে?
6. Branch separation আছে?
7. Offline sync plan আছে?
8. Docker deployment আছে?

9. Electron support আছে?
10. PWA support আছে?
11. Security architecture আছে?
12. License system আছে?

যদি missing থাকে:

Regenerate the architecture with the missing enterprise components.

PHASE 1 — CORE FOUNDATION

Goal

Core system foundation build করা

Build Order

1. Django project setup
2. PostgreSQL configuration
3. Redis configuration
4. Celery setup
5. Docker setup
6. Environment configuration
7. Base apps structure
8. Shared utilities
9. Logging system
10. Error handling
11. Health-check system
12. Audit log foundation

Prompt

Now generate Phase 1 implementation:

1. Django production project setup
2. PostgreSQL configuration
3. Redis + Celery setup
4. Docker configuration
5. Environment variable system
6. Logging system

7. Global error handling
8. Base reusable architecture
9. Shared utilities
10. Audit log foundation

Include:

- production folder structure
- complete files
- setup commands
- migration commands
- run instructions
- testing instructions
- security practices
- Docker run commands

Do not use placeholder code.

AFTER PHASE 1

You Must Do

1. VS Code এ open করুন
2. Docker run করুন
3. PostgreSQL connect check করুন
4. Redis check করুন
5. Celery check করুন
6. Migration run করুন
7. Server start করুন
8. Error fix করুন

Then Tell Claude

Phase 1 completed successfully. Now continue to Phase 2.

PHASE 2 — AUTHENTICATION + TENANTS + BRANCHES

Goal

Multi-company SaaS foundation

Features

1. JWT authentication
2. Refresh tokens
3. Role system
4. Permission system
5. Tenant system
6. Company system
7. Branch system
8. User management
9. Super admin panel
10. Audit logs
11. Login tracking
12. Device session tracking

Prompt

Now generate Phase 2 implementation:

1. JWT authentication
2. Refresh token system
3. Multi-tenant SaaS architecture
4. Company system
5. Branch system
6. Role-based access control
7. Permission matrix
8. User management
9. Device session tracking
10. Login history
11. Audit logging
12. Super admin controls

Include:

- database models
- migrations

- serializers
- APIs
- permissions
- middleware
- security validation
- test instructions
- API examples

Use production-grade architecture only.

PHASE 3 — LICENSE SYSTEM

Features

1. License generator
2. Activation server
3. Device binding
4. Subscription plans
5. Expiry system
6. Trial licenses
7. Lifetime licenses
8. Module locking
9. Remote deactivate
10. License analytics

Prompt

Now generate Phase 3 implementation:

1. Enterprise license system
2. Activation server
3. Device binding
4. Hardware fingerprint
5. Subscription plans
6. License expiry system
7. Trial management
8. Module-based licensing
9. Remote deactivation
10. License analytics

Include:

- secure encryption

- anti-tamper validation
 - activation APIs
 - database structure
 - admin controls
 - testing instructions
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-

PHASE 4 — INVENTORY CORE

Features

1. Products
2. Categories
3. Brands
4. Units
5. Warehouses
6. Branch stock
7. Stock movement
8. Stock ledger
9. Barcode
10. Variants
11. Batch tracking
12. Serial tracking
13. Damage stock
14. Stock transfer
15. Reorder alerts

Prompt

Now generate Phase 4 implementation:

1. Product management
2. Warehouse management
3. Branch inventory
4. Stock ledger
5. Inventory transactions
6. Barcode system
7. Product variants
8. Batch/serial tracking
9. Stock transfer
10. Damage stock handling
11. Inventory audit trail
12. Reorder alerts

Inventory calculations must be transaction-safe. Do not use fake logic.

PHASE 5 — PURCHASE SYSTEM

Features

1. Suppliers
2. Purchase orders
3. GRN
4. Purchase returns
5. Supplier ledger
6. Procurement workflow
7. RFQ system
8. Vendor comparison

Prompt

Now generate Phase 5 implementation:

1. Supplier management
2. Purchase orders
3. Goods receiving
4. Purchase returns
5. Supplier ledger
6. Procurement workflow
7. RFQ system
8. Approval workflow

Integrate fully with inventory and accounting.

PHASE 6 — SALES + POS

Features

1. POS
2. Sales invoices
3. Quotations
4. Returns
5. Customer ledger

6. Discounts
7. Loyalty system
8. Salesman tracking
9. Route sales
10. Thermal printing

Prompt

Now generate Phase 6 implementation:

1. POS system
2. Sales invoices
3. Customer management
4. Sales returns
5. Discount engine
6. Loyalty system
7. Salesman workflows
8. Thermal printing
9. Receipt printing
10. Offline sales queue

Integrate inventory deduction and accounting automatically.

PHASE 7 — ACCOUNTING

Features

1. Chart of accounts
2. Double-entry accounting
3. Journal entries
4. Cashbook
5. Bank
6. Expenses
7. Trial balance
8. Profit/Loss
9. Balance sheet
10. Tax/VAT

Prompt

Now generate Phase 7 implementation:

1. Double-entry accounting
2. Chart of accounts
3. Journal engine
4. Cashbook
5. Bank management
6. Expense management
7. Trial balance
8. Profit/Loss
9. Balance sheet
10. VAT/Tax system

Accounting logic must be fully transaction-safe.

PHASE 8 — OFFLINE + PWA

Features

1. IndexedDB
2. Service Worker
3. Offline queue
4. Sync engine
5. Conflict resolution
6. Background sync
7. Offline invoices
8. Offline inventory

Prompt

Now generate Phase 8 implementation:

1. PWA setup
2. Service Worker
3. IndexedDB storage
4. Offline-first architecture
5. Sync engine
6. Conflict resolution
7. Background synchronization
8. Offline inventory and invoice support

Include complete sync flow.

PHASE 9 — ELECTRON DESKTOP

Features

1. Electron app
2. SQLite offline DB
3. Auto updates
4. Desktop sync
5. Local printing
6. File sync

Prompt

Now generate Phase 9 implementation:

1. Electron desktop app
2. SQLite local database
3. Offline desktop mode
4. Auto-update system
5. Local synchronization
6. Secure desktop packaging

Generate production-ready Electron structure.

PHASE 10 — HRM + PAYROLL

Features

1. Employees
2. Attendance
3. Payroll
4. Leave
5. Overtime
6. Shifts
7. Employee documents

Prompt

Now generate Phase 10 implementation:

1. HRM system
2. Attendance tracking
3. Payroll engine
4. Leave management
5. Shift management
6. Overtime calculation
7. Employee records

Integrate accounting entries automatically.

PHASE 11 — REPORTS + DASHBOARDS

Features

1. Analytics
2. KPI dashboards
3. Reports
4. Graphs
5. PDF export
6. Excel export

Prompt

Now generate Phase 11 implementation:

1. Analytics dashboard
 2. KPI widgets
 3. Report engine
 4. PDF export
 5. Excel export
 6. Real-time graphs
 7. Executive dashboards
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-

PHASE 12 — NOTIFICATIONS + COMMUNICATION

Features

1. Email
2. SMS
3. Push notifications
4. WhatsApp-ready
5. Internal notifications

Prompt

Now generate Phase 12 implementation:

1. Email notification system
 2. SMS gateway integration
 3. Push notification engine
 4. Internal notification center
 5. Real-time alerts
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PHASE 13 — BACKUP + SECURITY

Features

1. Backup
2. Restore
3. Encryption
4. Audit
5. Security logs
6. Disaster recovery

Prompt

Now generate Phase 13 implementation:

1. Backup system
2. Restore system
3. Encrypted backups
4. Disaster recovery

5. Security monitoring
 6. Audit center
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PHASE 14 — DEPLOYMENT

Features

1. Docker
2. Nginx
3. Gunicorn
4. SSL
5. CI/CD
6. VPS deployment

Prompt

Now generate Phase 14 implementation:

1. Docker deployment
 2. Nginx configuration
 3. Gunicorn setup
 4. SSL configuration
 5. CI/CD pipeline
 6. VPS deployment guide
 7. Production environment setup
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PHASE 15 — FINAL HARDENING

Final Checks

1. Load testing
2. Security testing
3. Penetration testing
4. API validation
5. Transaction validation
6. Offline sync validation
7. Multi-branch validation
8. Backup recovery validation

9. Accounting integrity validation
10. License validation

Prompt

Now perform enterprise hardening review.

Generate: 1. Security audit checklist 2. Performance optimization checklist 3. Load testing strategy 4. Database optimization strategy 5. Backup validation strategy 6. Sync validation strategy 7. Production deployment checklist 8. Enterprise QA checklist

IMPORTANT RULES FOR EVERY PHASE

Always ask Claude for:

1. Complete files
 2. Folder structure
 3. Setup commands
 4. Migration commands
 5. Run instructions
 6. API examples
 7. Testing instructions
 8. Security validation
 9. Edge-case handling
 10. Integration flow
-

HOW TO FOLLOW THIS ROADMAP (MOST IMPORTANT)

এই roadmap এ দুই ধরনের numbering আছে:

TYPE 1 — STAGE 0 → STAGE 16

এটাই আসল build order ।

আপনি ERP build করার সময় সবসময়:

STAGE 0 তারপর STAGE 1 তারপর STAGE 2 এভাবে serial অনুযায়ী কাজ করবেন ।

কখনো stage skip করবেন না।

কারণ:

- আগে database ছাড়া inventory build করলে future এ rewrite লাগবে
- auth ছাড়া dashboard build করলে security ভাঙবে
- accounting ছাড়া sales build করলে transaction mismatch হবে
- sync ছাড়া offline system unstable হবে

তাই:

✓ সবসময় STAGE serial follow করবেন ✗ কখনো random module build করবেন না ✗ কখনো UI আগে full build করবেন না ✗
কখনো accounting পরে চিন্তা করবেন না

REAL BUILD FLOW (VERY IMPORTANT)

FIRST

STAGE 0 → Planning + Architecture

SECOND

STAGE 1 → Backend Foundation

THIRD

STAGE 2 → Authentication + Security

FOURTH

STAGE 3 → SaaS + Branch + License

FIFTH

STAGE 4 → Database + Offline + Sync Architecture

SIXTH

STAGE 5 → Inventory System

SEVENTH

STAGE 6 → Purchase System

EIGHTH

STAGE 7 → Sales + POS

NINTH

STAGE 8 → Accounting System

TENTH

STAGE 9 → HRM + CRM

ELEVENTH

STAGE 10 → Reports + Analytics

TWELFTH

STAGE 11 → Offline + Realtime System

THIRTEENTH

STAGE 12 → Electron Desktop App

FOURTEENTH

STAGE 13 → Enterprise Extra Features

FIFTEENTH

STAGE 14 → Security + Backup + Recovery

SIXTEENTH

STAGE 15 → Deployment + Scaling

FINAL

STAGE 16 → Final Testing + Hardening

TYPE 2 — FEATURE INDEX (1 → 66)

এই numbering build order ন।

এগুলো শুধু:

- feature categories
- enterprise checklist
- future expansion list

এই list দিয়ে শুধু check করবেন: সব feature আছে কিনা।

কাজ করবেন সবসময়: STAGE serial অনুযায়ী।

COMPLETE MASTER BUILD ORDER (0 → 66)

এই section সবচেয়ে important। এখানে exactly বলা আছে:

- কোনটার পরে কোনটা build করবেন
 - কোন module আগে build করলে safe হবে
 - কোনটা পরে করলে rewrite কন হবে
 - কোনটা ছাড়া পরের module build করা উচিত না
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STAGE 0 — MASTER PLANNING

0.1 Business Planning

1. ERP business model
2. SaaS pricing model
3. License structure
4. Subscription plans
5. Company branding
6. Version naming system
7. Customer target planning
8. Module planning
9. Deployment planning
10. Support system planning

0.2 Enterprise Architecture Planning

1. Folder architecture
2. Backend architecture
3. Frontend architecture
4. API architecture
5. Offline architecture
6. Electron architecture
7. Multi-tenant architecture

8. Branch architecture
9. Inventory architecture
10. Accounting architecture

0.3 Database Planning

1. PostgreSQL schema planning
 2. Tenant isolation planning
 3. Branch separation planning
 4. Inventory table planning
 5. Accounting table planning
 6. Audit table planning
 7. Sync table planning
 8. Backup planning
 9. Index planning
 10. Scalability planning
-

UI/UX MASTER DESIGN REQUIREMENTS

Loading Screen (Splash Screen)

Purpose

Software open হওয়ার সময় premium enterprise feel দিতে হবে।

Loading Screen Requirements

1. S4 Business Thinking logo animation
2. Animated loading progress
3. Version display (Example: S4-143)
4. Secure initialization checking
5. Database connection checking
6. License validation checking
7. Offline/online status checking
8. Smooth fade animation
9. Dark/light mode support
10. Mobile + desktop responsive
11. Fast optimized loading
12. PWA startup support
13. Electron startup support
14. Brand color animation
15. Modern enterprise UI feel

Loading Screen Design Style

1. Clean premium UI
 2. Enterprise dashboard style
 3. Smooth animations
 4. Modern glassmorphism support
 5. Professional typography
 6. High-end SaaS appearance
 7. Minimal but powerful design
 8. GPU-optimized animations
-

Login System UI/UX

Login Page Requirements

1. Modern enterprise login UI
2. Mobile-first responsive design
3. Company branding support
4. Branch/company selector
5. Email/phone/username login
6. Password visibility toggle
7. Remember device option
8. Secure JWT login
9. OTP-ready architecture
10. Multi-language support
11. Dark/light mode
12. Offline login support
13. Device binding support
14. Login activity tracking
15. Animated transitions

Advanced Login Features

1. Forgot password system
2. Email verification
3. Login session management
4. Device management
5. Account lock protection
6. Failed login protection
7. Captcha-ready support
8. Multi-device login control
9. Super admin login
10. Tenant/company login routing

Login UI Design Style

1. Premium SaaS style

2. Enterprise-grade layout
 3. Glassmorphism support
 4. Smooth animations
 5. Professional dashboard preview
 6. Responsive mobile layout
 7. Tablet optimization
 8. Desktop optimization
 9. Clean typography
 10. Fast-loading optimized UI
-

Main Dashboard UI/UX

Dashboard Requirements

1. Real-time widgets
2. KPI cards
3. Sales analytics
4. Inventory alerts
5. Notification center
6. Activity timeline
7. Branch switching
8. Quick actions panel
9. AI insights panel
10. Mobile dashboard optimization

Dashboard Design Style

1. Modern enterprise SaaS UI
 2. Responsive grid system
 3. Drag-drop widgets
 4. Glassmorphism cards
 5. Smooth transitions
 6. Realtime animations
 7. Clean navigation system
 8. Sidebar + topbar layout
 9. Mobile navigation drawer
 10. Premium visual hierarchy
-

STAGE 1 — FOUNDATION

1.1 Core Backend Setup

1. Django setup
2. DRF setup

3. FastAPI setup
4. PostgreSQL setup
5. Redis setup
6. Celery setup
7. Environment configuration
8. Docker setup
9. Logging system
10. Error handling system

1.2 Base Infrastructure

1. Shared utilities
 2. Common services
 3. Reusable modules
 4. Base models
 5. Base APIs
 6. Global exception handling
 7. API response structure
 8. Health check APIs
 9. Monitoring setup
 10. Audit foundation
-

STAGE 2 — AUTH + SECURITY

2.1 Authentication

1. JWT authentication
2. Refresh token system
3. Secure login system
4. Password reset system
5. Device login tracking
6. Session management
7. Multi-device login
8. OTP ready structure
9. Email verification
10. Login history

2.2 Enterprise Security

1. RBAC
2. Permission matrix
3. API security
4. Rate limiting
5. CSRF protection
6. XSS protection

7. SQL injection prevention
 8. Audit logs
 9. Secure headers
 10. Activity monitoring
-

STAGE 3 — SAAS + TENANTS + LICENSE

3.1 Multi-Tenant System

1. Tenant system
2. Company system
3. Branch system
4. Warehouse system
5. Tenant isolation
6. Branch isolation
7. Super admin controls
8. Tenant analytics
9. Resource isolation
10. Usage tracking

3.2 License System

1. License generator
 2. Activation server
 3. Device binding
 4. Hardware lock
 5. Trial system
 6. Subscription system
 7. Expiry system
 8. Module locking
 9. Remote deactivate
 10. License analytics
-

STAGE 4 — DATA ARCHITECTURE

4.1 Production Database

1. PostgreSQL production DB
2. ACID transactions
3. High-performance indexing
4. Read/write optimization
5. Replication-ready structure

6. Backup-ready structure
7. High-availability planning
8. Database migrations
9. Partition-ready tables
10. Transaction-safe design

4.2 Offline Database

1. SQLite offline DB
2. IndexedDB storage
3. Offline queue
4. Local sync cache
5. Local audit logs
6. Offline invoice storage
7. Offline inventory queue
8. Local backup storage
9. Offline encryption
10. Conflict-safe storage

4.3 Synchronization Engine

1. Real-time sync
2. Background sync
3. Queue-based sync
4. Conflict resolution
5. Failed sync retry
6. Partial sync
7. Multi-device sync
8. Branch sync
9. Version tracking
10. Sync logs

STAGE 5 — INVENTORY CORE

5.1 Inventory Base

1. Products
2. Categories
3. Brands
4. Units
5. Warehouses
6. Product variants
7. Barcode system
8. QR system
9. Batch tracking

10. Serial tracking

5.2 Inventory Transactions

1. Stock ledger
 2. Stock movement
 3. Branch stock
 4. Stock transfer
 5. Damage stock
 6. Adjustment system
 7. FIFO/LIFO
 8. Reorder alerts
 9. Landed cost
 10. Inventory audit trail
-

STAGE 6 — PURCHASE + SUPPLY CHAIN

1. Supplier management
 2. Purchase orders
 3. GRN system
 4. Purchase returns
 5. RFQ system
 6. Vendor comparison
 7. Procurement workflow
 8. Approval workflow
 9. Supplier ledger
 10. Supply chain tracking
-

STAGE 7 — SALES + POS

1. POS system
 2. Sales invoices
 3. Quotations
 4. Sales returns
 5. Customer ledger
 6. Loyalty system
 7. Discount engine
 8. Salesman tracking
 9. Route sales
 10. Thermal printing
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STAGE 8 — ACCOUNTING

1. Double-entry accounting
 2. Journal engine
 3. Chart of accounts
 4. Cashbook
 5. Bank system
 6. Expense management
 7. Trial balance
 8. Profit/Loss
 9. Balance sheet
 10. VAT/Tax
-

STAGE 9 — HRM + CRM

1. HRM
 2. Payroll
 3. Attendance
 4. Leave management
 5. CRM
 6. Lead management
 7. Customer history
 8. Ticket/support system
 9. Internal messaging
 10. Employee tracking
-

STAGE 10 — REPORTS + ANALYTICS

1. Dashboard
 2. KPI system
 3. Analytics engine
 4. PDF export
 5. Excel export
 6. BI reports
 7. Forecasting
 8. Executive dashboard
 9. Audit reports
 10. Custom report builder
-

STAGE 11 — OFFLINE + REALTIME

1. PWA
 2. Service Worker
 3. IndexedDB
 4. Offline-first architecture
 5. WebSocket
 6. Real-time notifications
 7. Conflict resolution
 8. Offline invoice queue
 9. Offline inventory sync
 10. Background synchronization
-

STAGE 12 — ELECTRON DESKTOP

1. Electron app
 2. SQLite integration
 3. Auto updates
 4. Desktop sync
 5. Local printing
 6. Offline desktop mode
 7. File synchronization
 8. Secure packaging
 9. Installer system
 10. Update rollback
-

STAGE 13 — ENTERPRISE FEATURES

1. AI system
 2. Automation engine
 3. Workflow builder
 4. IoT integration
 5. Smart hardware support
 6. Marketplace system
 7. Plugin system
 8. White-label system
 9. Mobile app APIs
 10. Global expansion features
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STAGE 14 — SECURITY + BACKUP

1. Backup system
 2. Restore system
 3. Disaster recovery
 4. Encrypted backups
 5. Security monitoring
 6. Compliance logs
 7. Immutable audit logs
 8. Tamper detection
 9. Penetration testing
 10. Recovery testing
-

STAGE 15 — DEPLOYMENT + SCALING

1. Docker deployment
 2. Nginx
 3. Gunicorn
 4. SSL
 5. VPS deployment
 6. CI/CD
 7. Load balancing
 8. Auto scaling
 9. High availability
 10. Production monitoring
-

STAGE 16 — FINAL HARDENING

1. Load testing
 2. Stress testing
 3. Security testing
 4. API testing
 5. Offline sync testing
 6. Inventory validation
 7. Accounting validation
 8. Backup validation
 9. Multi-branch validation
 10. Production readiness validation
-

REAL ENTERPRISE BUILD ORDER

STAGE 1 — FOUNDATION

1. Project Architecture

Purpose:

- Define scalable enterprise structure
- Prevent future rewrite problems
- Prepare backend/frontend/API structure

Build: 1. Backend architecture 2. Frontend architecture 3. API structure 4. Modular folder structure 5. SaaS structure 6. Multi-tenant structure 7. Multi-branch structure 8. Shared utilities structure 9. Logging structure 10. Deployment structure

Claude Prompt:

Design the complete production-grade ERP architecture first. Generate: 1. Folder structure 2. Backend architecture 3. Frontend architecture 4. API architecture 5. Multi-tenant architecture 6. Branch architecture 7. Deployment architecture 8. Offline-online architecture 9. Security architecture 10. License architecture

Then wait for approval.

2. Database Architecture

Purpose:

- Prevent inventory corruption
- Prevent accounting mismatch
- Prevent future database rewrite

Build: 1. PostgreSQL schema 2. Tenant isolation 3. Branch separation 4. Inventory tables 5. Accounting tables 6. Audit tables 7. License tables 8. Sync tables 9. User/session tables 10. Reporting tables

Claude Prompt:

Generate enterprise-grade PostgreSQL database architecture. Include: 1. Multi-tenant schemas 2. Inventory relationships 3. Accounting relationships 4. Branch relationships 5. Audit relationships 6. Sync tables 7. Optimized indexes 8. Transaction-safe design 9. Backup strategy 10. High-scale optimization

3. Authentication + Security

Purpose:

- Every module depends on users and permissions

Build: 1. JWT authentication 2. Refresh tokens 3. RBAC 4. Session management 5. Device tracking 6. Audit logs 7. Security middleware 8. Rate limiting 9. API security 10. Secure password system

4. Multi-Tenant + Branch System

Purpose:

- SaaS ERP foundation

Build: 1. Company system 2. Branch system 3. Tenant isolation 4. Branch permissions 5. Tenant filtering 6. Super admin controls 7. Subscription linking 8. Branch inventory separation 9. Branch accounting separation 10. Tenant analytics

5. License System

Purpose:

- Commercial ERP selling foundation

Build: 1. License activation 2. Subscription plans 3. Device lock 4. Expiry system 5. Trial licenses 6. Module locking 7. License analytics 8. Remote deactivate 9. Offline activation 10. Anti-tamper protection

STAGE 2 — CORE BUSINESS ENGINE

6. Inventory Core

Build: 1. Products 2. Categories 3. Warehouses 4. Branch stock 5. Stock ledger 6. Stock movement 7. Barcode 8. Variants 9. Batch tracking 10. Serial tracking

7. Purchase System

Build: 1. Suppliers 2. Purchase orders 3. Goods receiving 4. Purchase returns 5. Procurement workflow 6. Supplier ledger 7. Approval workflow 8. Vendor comparison 9. Cost tracking 10. Inventory integration

8. Sales + POS

Build: 1. POS 2. Sales invoices 3. Quotations 4. Sales returns 5. Customer ledger 6. Discounts 7. Loyalty system 8. Thermal printing 9. Offline sales queue 10. Accounting integration

9. Accounting Engine

Build: 1. Double-entry accounting 2. Journal entries 3. Chart of accounts 4. Cashbook 5. Bank management 6. Expenses 7. Trial balance 8. Profit/Loss 9. Balance sheet 10. VAT/Tax system

STAGE 3 — ADVANCED OPERATIONS

10. HRM + Payroll

11. CRM + Customer System

12. Notifications + Messaging

13. Reporting + Analytics

14. AI + Forecasting

STAGE 4 — OFFLINE + REALTIME

15. PWA + IndexedDB

16. Offline Sync Engine

17. Conflict Resolution System

18. WebSocket Realtime System

19. Electron Desktop App

STAGE 5 — ENTERPRISE HARDENING

20. Backup + Recovery

21. Audit + Monitoring

22. Security Hardening

23. Load Testing

24. Docker + VPS Deployment

25. CI/CD + Auto Updates

STAGE 6 — FUTURE ECOSYSTEM

26. Marketplace

27. Plugin System

28. API Ecosystem

29. White Label System

30. Mobile Apps

এখনো **missing** আছে কিনা — Final Enterprise Checklist

Core ERP

- Inventory
- Purchase
- Sales
- POS
- Accounting
- HRM
- Payroll
- CRM
- Reports
- Analytics
- Notifications
- Audit Logs
- Permissions

SaaS System

- Multi-tenant
- Multi-company
- Multi-branch
- License system
- Subscription system
- Super admin panel

Offline + Desktop

- PWA
- IndexedDB
- Service Worker
- Offline sync
- Conflict resolution
- Electron desktop app
- SQLite offline database

Enterprise Security

- JWT
- Refresh token
- RBAC
- Device tracking
- API security
- Audit logs
- Backup
- Disaster recovery

Infrastructure

- PostgreSQL
- Redis
- Celery
- Docker
- Nginx
- Gunicorn
- CI/CD
- VPS deployment

Future Expansion

- Marketplace
- Plugin system
- Mobile apps
- AI support

- IoT support
 - White-label system
-

MASTER ENTERPRISE FEATURE INDEX (1 → 66)

- 1. Financial & Accounting**
- 2. Inventory Advanced**
- 3. CRM & Communication**
- 4. Ecommerce & Integration**
- 5. Backup & Recovery**
- 6. Monitoring & Maintenance**
- 7. AI & Smart Features**
- 8. User Experience**
- 9. Mobile & Desktop Features**
- 10. Compliance & Legal**
- 11. Super Admin SaaS Panel**
- 12. File & Document System**
- 13. Development & DevOps**
- 14. Project Architecture**
- 15. Authentication & Security**
- 16. Multi-Tenant SaaS**
- 17. Branch Management**
- 18. Warehouse Management**
- 19. Product Management**

- 20. Inventory Ledger System**
- 21. Barcode & QR System**
- 22. Purchase Management**
- 23. Supplier Management**
- 24. Sales Management**
- 25. POS System**
- 26. Customer Management**
- 27. Advanced Accounting & Finance**
- 28. Procurement & Supply Chain**
- 29. Advanced Sales System**
- 30. Enterprise Workflow Automation**
- 31. Document & Digital Signature System**
- 32. Enterprise Search Engine**
- 33. Localization & Internationalization**
- 34. Education & Training System**
- 35. AI Automation & Intelligence**
- 36. Franchise & Chain Business Management**
- 37. Hospitality & Restaurant Extensions**
- 38. Marketplace & Vendor Ecosystem**
- 39. Disaster Recovery & High Availability**

- 40. Enterprise Administration System**
- 41. Advanced Audit & Compliance**
- 42. Enterprise Notification System**
- 43. Advanced Access Control**
- 44. Data Intelligence & Warehousing**
- 45. Enterprise Messaging & Collaboration**
- 46. Scheduling & Calendar System**
- 47. Enterprise Printing System**
- 48. Smart Automation Engine**
- 49. Cloud & Hybrid Infrastructure**
- 50. Enterprise Customization System**
- 51. ERP Ecosystem Expansion**
- 52. Advanced Technical Architecture**
- 53. Quality Assurance & Stability**
- 54. Enterprise Update System**
- 55. Advanced Media & Storage System**
- 56. Advanced AI & Automation Ecosystem**
- 57. IoT & Smart Hardware Ecosystem**
- 58. Enterprise Communication Hub**
- 59. Global Enterprise Expansion**

60. Government & Compliance Integration

61. Advanced Business Intelligence

62. Enterprise Data Protection

63. Enterprise Marketplace & Ecosystem

64. Enterprise Operational Control

65. Future Technology Readiness

66. Enterprise Data Architecture

Final Reality Check

এখন feature side থেকে বড় কোনো enterprise-level feature missing নাই।

এখন সবচেয়ে important হবে:

1. Proper database design
2. Safe inventory transactions
3. Safe accounting logic
4. Stable offline sync
5. Proper API structure
6. Good testing
7. Performance optimization
8. Security hardening
9. Backup validation
10. Load testing

কারণ real ERP এ সবচেয়ে বড় challenge:

- stable system
 - data corruption না হওয়া
 - accounting mismatch না হওয়া
 - offline sync safe রাখা
 - thousands user handle করা
-

FINAL RESULT

এই roadmap follow করলে আপনি build করতে পারবেন:

1. Enterprise ERP
2. SaaS platform
3. Online + Offline system
4. Desktop ERP
5. Multi-branch ERP
6. Multi-company ERP
7. License-controlled ERP
8. PWA ERP
9. High-scale ERP
10. Commercial sellable ERP ecosystem